Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-F Pericardial Disease

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-06-C01-BD1

Brain dump 1 · foundational retrieval: Pericardial layers and the findings of acute pericarditis

A competent response contains

· The pericardium has an outer fibrous sac of dense collagen and an inner serous sac; the parietal serous layer lines the fibrous sac and the visceral layer, the epicardium, covers the heart.

· Between the layers lies about 15 to 50 mL of serous fluid, a plasma ultrafiltrate that lets the heart move with little friction.

· The fibrous and parietal layers receive sensory fibers from the phrenic nerves (C3 to C5); the visceral layer is insensitive to pain.

· Pain is sharp and pleuritic because the inflamed parietal layer is stretched and rubbed with each heartbeat and breath; it worsens lying flat and with deep inspiration and eases on sitting up and leaning forward, which is thought to reduce pressure on the inflamed parietal surface.

· Phrenic innervation refers the pain to the trapezius ridge, a feature ischemic pain lacks.

· The friction rub is fibrin-coated layers grating together: scratchy, up to three components (atrial systole, ventricular systole, early diastolic filling), best heard at the lower left sternal border with the patient leaning forward, and it comes and goes.

· Low-grade fever, leukocytosis, and raised C-reactive protein reflect systemic inflammation.

· Most cases in developed countries are labeled idiopathic and presumed viral; coxsackie B virus is the classic cause, with echovirus, adenovirus, influenza, and HIV as others, in young and middle-aged adults, more often men.

· Raised vascular permeability lets fibrinogen escape and polymerize on both serous surfaces, giving a fibrinous or serofibrinous exudate that usually resolves without scarring.

Distinctions that matter

· Parietal and fibrous layers carry pain; the visceral layer does not, so the pain comes from the outer layers being rubbed.

· A rub can persist even once some fluid has separated the layers, so a rub and an effusion are not mutually exclusive.

Common misconceptions

· Saying the serous fluid is secreted. It is a plasma ultrafiltrate, which is why Starling forces alone can enlarge it.

· Attributing the trapezius pain to the heart itself. It follows the phrenic nerve supply to the fibrous and parietal layers.

Where it is taught: Section C01, first three paragraphs of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C01-BD2

Brain dump 2 · integration and application: Read the tracing back to the inflamed epicardium

A competent response contains

· The pericardium generates no electrical signal; the changes come from inflammation of the superficial myocardium beneath the visceral layer.

· Injured epicardial cells create a current of injury that elevates the ST segment in leads facing the epicardium.

· Because the whole epicardial surface is involved, elevation appears in most limb and precordial leads rather than in one coronary territory.

· Diffuse epicardial injury has no opposite wall to show reciprocal change, so there is no reciprocal depression outside aVR.

· Lead aVR looks into the cavities from above, so it shows the mirror image: ST depression and PR elevation.

· PR depression comes from atrial epicardial injury displacing the PR segment, so the atrial epicardium is involved as well.

· Evolution in order: diffuse ST elevation, then ST returns to baseline with T flattening, then T inversion, then normalization, over days to weeks.

· A troponin rise means the adjacent myocardium is inflamed, which is myopericarditis.

Distinctions that matter

· ST elevation here reports epicardial inflammation, not coronary occlusion, which is why it crosses territories.

· In pericarditis T inversion appears after the ST returns to baseline, while in infarction it often appears while the ST is still elevated.

Common misconceptions

· Expecting reciprocal depression in the opposite leads. With the whole surface injured there is no uninvolved wall to show it.

· Reading PR depression as a conduction problem. It is atrial epicardial injury displacing the segment.

Where it is taught: Section C01, electrocardiogram paragraph, the comparison table, and the stop and predict box of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C01-CP01

Comparison table: Acute pericarditis against ST elevation myocardial infarction

A competent response contains

· Distribution: pericarditis elevates ST diffusely in most leads; infarction elevates ST in one coronary territory.

· Shape: concave upward in pericarditis, often convex or straight in infarction.

· Reciprocal depression: only aVR and sometimes V1 in pericarditis; in the opposite leads in infarction.

· PR segment: depressed in pericarditis and elevated in aVR; usually unchanged in infarction.

· Q waves: absent in pericarditis; they develop with necrosis in infarction.

· T wave inversion: after the ST returns to baseline in pericarditis; often while the ST is still elevated in infarction.

· The shift arises from injured epicardium in pericarditis, and because the injury covers the whole heart there is no opposite wall to show reciprocal change apart from leads looking into the cavity.

Distinctions that matter

· Lead grouping is the first discriminator: one territory points to a vessel, most leads point to a surface.

· A PR shift belongs to pericardial inflammation and is not part of the infarct pattern.

Common misconceptions

· Treating ST shape as decisive on its own. Shape supports the pattern but distribution, PR shift, and Q waves carry more weight.

· Expecting Q waves to develop in pericarditis. There is no transmural necrosis, so they do not appear.

Where it is taught: Section C01, the table of acute pericarditis versus ST elevation myocardial infarction of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C01-Q01

Board-style question: Diffuse ST elevation with PR depression

Correct answer: C. Current of injury from inflammation of the epicardial myocardium across the whole cardiac surface

Why it is right: Inflammation of the superficial myocardium beneath the visceral layer creates a current of injury across the entire epicardial surface, so the ST segment is elevated in most limb and precordial leads, PR depression appears from atrial epicardial injury, and only aVR, which looks into the cavities, shows the mirror image.

A is wrong: A dominant right coronary artery occlusion would elevate the ST segment in the inferior and lateral leads with reciprocal depression in the opposite leads, and it would not produce anterior elevation with PR depression in every affected lead.

B is wrong: The fibrous and parietal pericardial layers generate no electrical signal, which is why the tracing reflects the underlying myocardium rather than the inflamed sac itself.

D is wrong: Beat to beat change in cardiac axis within pericardial fluid is electrical alternans, which alters QRS amplitude or axis on alternate beats; fluid between the heart and the electrodes lowers QRS voltage rather than elevating the ST segment.

A competent response contains

· Diffuse concave ST elevation with PR depression and the mirror image in aVR is acute pericarditis, most often idiopathic and presumed viral after a respiratory or gastrointestinal illness.

· The pericardium is electrically silent, so the ST shift comes from a current of injury in the epicardial myocardium under the visceral layer, which is why it spans coronary territories.

Distinctions that matter

· A coronary occlusion injures the wall supplied by one vessel; epicardial inflammation injures the whole surface.

Common misconceptions

· Looking for a single vessel to explain elevation in inferior, lateral, and anterior leads at once.

Where it is taught: Section C01 of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C02-BD1

Brain dump 1 · foundational retrieval: Build the secondary pericarditis table from memory

A competent response contains

· The exudate reflects how strongly and in what way the outside disease injures the serosa: mild permeability gives serous fluid, greater permeability lets fibrinogen escape and coat the layers, neutrophil-recruiting organisms give pus, damaged or fragile vessels and bleeding tendency add blood, and granulomatous infection gives caseous necrosis.

· The shaggy surface of severe fibrinous pericarditis is described as bread and butter.

· Early post-infarction pericarditis, about 1 to 3 days: necrosis of a transmural infarct inflames the overlying pericardium, giving fibrinous or fibrinohemorrhagic exudate over the infarct.

· Dressler syndrome, weeks after infarction: autoimmune reaction to released cardiac antigens, fibrinous, often with effusion.

· Cardiac surgery: direct trauma, and later a Dressler-like post-pericardiotomy syndrome, fibrinous or hemorrhagic.

· Uremia: retained metabolic toxins injure the serosa, fibrinous, and hemorrhagic with uremic platelet dysfunction.

· Systemic lupus erythematosus: immune complex deposition, serous or fibrinous. Rheumatoid arthritis: immune complexes and rheumatoid inflammation, fibrinous, may later constrict.

· Mediastinal radiation: endothelial and serosal injury, fibrinous acutely with fibrosis later.

· Tuberculosis: granulomatous response to Mycobacterium tuberculosis, caseous and often hemorrhagic. Pyogenic bacteria: spread from pneumonia or empyema, blood, or surgery, purulent. Cancer: tumor implants with fragile vessels and blocked lymphatics, hemorrhagic.

· Fibrinous and serofibrinous exudates are the most common; purulent and caseous exudates are the most likely to organize into constriction.

Distinctions that matter

· Immune complex disease gives serous or fibrinous fluid, while organisms that recruit neutrophils give pus, so the cell response rather than the severity decides between them.

· Hemorrhagic fluid has several routes: vessel damage, fragile tumor vessels, and a bleeding tendency such as uremic platelet dysfunction.

Common misconceptions

· Treating every post-infarction pericarditis as one entity. Timing separates direct inflammation at 1 to 3 days from the immune reaction weeks later.

· Expecting uremic pericarditis to be infectious. It comes from retained toxins.

Where it is taught: Section C02, opening paragraph and the table of causes and their usual exudate of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C02-BD2

Brain dump 2 · integration and application: Four patients, four exudates

A competent response contains

· Patient 1: early post-infarction pericarditis, fibrinous or fibrinohemorrhagic over the infarct, identified by the day 2 timing after a transmural infarct, localized and self-limited.

· Patient 1 would not be expected to have pericarditis with a subendocardial infarct, because the necrosis has to reach the epicardium to inflame the overlying pericardium.

· Patient 2: Dressler syndrome, fibrinous with effusion, identified by the delay of weeks plus fever and pleuritis; the delay fits priming of antibodies and T cells against released cardiac antigens rather than direct injury, and the same picture after cardiac surgery is post-pericardiotomy syndrome.

· Patient 3: uremic pericarditis from retained metabolic toxins, fibrinous and hemorrhagic; diffuse ST elevation is often absent because the inflammation tends not to involve the epicardial myocardium, and uremic platelet dysfunction makes the fluid bloody and tamponade more likely.

· Patient 4: bacterial pericarditis, usually Staphylococcus aureus or Streptococcus pneumoniae, purulent, reaching the sac by spread from pneumonia or empyema, acute and toxic.

· Patient 4 is the most likely of these to heal by dense fibrosis, since purulent and caseous exudates organize into constriction, as does radiation injury.

Distinctions that matter

· Early post-infarction pericarditis is a local response to necrosis; Dressler syndrome is systemic and immune, with fever and pleuritis as well.

· Absence of diffuse ST elevation does not exclude pericarditis, because the tracing depends on epicardial myocardial involvement rather than on the sac.

Common misconceptions

· Relying on the electrocardiogram to rule out pericarditis in renal failure.

· Expecting a purulent exudate to resolve cleanly like a viral fibrinous one. Pus organizes and can constrict.

Where it is taught: Section C02, post-infarction paragraph, the Dressler flow, and the final paragraph on uremia, bacteria, and tuberculosis of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C02-RT01

Retrieval exercise: Rebuild the Dressler syndrome sequence

A competent response contains

· Step 1: infarction or surgery injures myocardium and pericardium.

· Step 2: hidden myocardial antigens are released; they normally sit inside myocytes, so the immune system has not been exposed to them.

· Step 3: antibodies and T cells are primed over weeks, which is an adaptive response and explains the delay.

· Step 4: immune attack on pericardium and pleura.

· Step 5: fever, pleuritic pain, rub, and effusion appear weeks later, with a fibrinous exudate in the sac.

· The delay of weeks is the feature that separates Dressler syndrome from early post-infarction pericarditis at 1 to 3 days.

· The same process after cardiac surgery is called post-pericardiotomy syndrome.

Distinctions that matter

· Direct inflammation from necrosis is immediate and local; an adaptive immune response needs weeks and is systemic.

· Pleural involvement belongs to the immune process, not to early post-infarction pericarditis.

Common misconceptions

· Calling Dressler syndrome a second infarct or reinfarction. The injury has already happened and the attack is immune.

· Expecting it within days. Priming takes weeks, usually given as about 2 to 10 weeks.

Where it is taught: Section C02, the flow on Dressler syndrome as an immune reaction to injury of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C03-BD1

Brain dump 1 · foundational retrieval: The rigid shell and what it does to filling

A competent response contains

· Sequence: fibrinous, purulent, or caseous exudate; granulation tissue organizes the exudate; visceral and parietal layers fuse; dense fibrosis, sometimes with calcification; a rigid shell fixes total cardiac volume.

· Tuberculosis is a major cause worldwide; in developed countries most cases follow idiopathic or viral pericarditis, cardiac surgery, or mediastinal radiation, for example for Hodgkin lymphoma.

· The shell fixes total cardiac volume but the myocardium inside relaxes normally.

· In early diastole the ventricles fill very rapidly, driven by high atrial pressure, until they reach the limit of the shell, where filling stops abruptly.

· The abrupt halt makes the pericardial knock, an early diastolic sound that comes earlier and is higher pitched than an S3.

· On ventricular pressure tracings the same event gives a dip-and-plateau or square root pattern, and the jugular venous pulse shows a steep y descent.

· Diastolic pressures in all four chambers rise and become nearly equal.

· Systolic function is preserved but limited filling lowers stroke volume; high venous pressure gives hepatomegaly, ascites, and edema out of proportion to lung congestion.

· Plain films or CT may show pericardial calcification.

Distinctions that matter

· Filling is not reduced throughout diastole: it is very rapid early and then stops, which is what the dip-and-plateau tracing records.

· The knock is earlier and higher pitched than an S3, which is how the two are told apart at the bedside.

Common misconceptions

· Assuming the myocardium must be stiff for filling to be limited. In constriction the muscle relaxes normally and the sac sets the limit.

· Expecting pulmonary congestion to dominate. Systemic venous signs are out of proportion to lung findings.

Where it is taught: Section C03, first two blocks and the paragraph on the fixed shell of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C03-BD2

Brain dump 2 · integration and application: Kussmaul sign and telling the sac from the muscle

A competent response contains

· Normally inspiration lowers intrathoracic pressure, the fall is transmitted to the right atrium, venous return is drawn in, and the neck veins fall.

· In constriction the jugular venous pressure rises or fails to fall with inspiration, which is Kussmaul sign.

· Mechanism: the rigid shell blocks transmission of intrathoracic pressure to the heart, while descent of the diaphragm still pushes abdominal venous blood toward a right heart that cannot expand, so venous pressure rises.

· Reciprocal filling: the pulmonary veins lie outside the shell, so inspiration lowers their pressure but not left heart pressure, left filling falls, and the extra right ventricular volume pushes the septum left.

· Kussmaul sign is not specific; it also occurs in restrictive cardiomyopathy and right ventricular infarction.

· Constriction against restriction: stiffness in the pericardium against the myocardium; pericardium thickened and possibly calcified against normal; myocardial relaxation normal against impaired; exaggerated reciprocal respiratory variation against little; left and right diastolic pressures nearly equal against left usually higher; pulmonary hypertension usually mild against often marked; natriuretic peptide normal or mildly raised, because the shell limits myocardial stretch, against markedly raised; pericardial knock against a possible S3; low voltage with nonspecific T changes against low voltage with conduction disease suggesting amyloid.

· Restrictive cardiomyopathy comes from a myocardium infiltrated or scarred by amyloid, sarcoidosis, hemochromatosis, or endomyocardial fibrosis, with a normal pericardium.

· Both limit filling with a preserved ejection fraction and both give dip-and-plateau tracings, so the location of the stiffness explains every difference.

Distinctions that matter

· Natriuretic peptide separates the two because the shell prevents the myocardial stretch that drives its release, while an infiltrated myocardium is stretched and releases a great deal.

· Respiratory variation needs a fixed shared space; with a compliant pericardium each stiff ventricle limits only its own filling.

Common misconceptions

· Taking Kussmaul sign as proof of constriction. It appears in restrictive cardiomyopathy and right ventricular infarction too.

· Expecting marked pulmonary hypertension in constriction. It is usually mild, and marked pulmonary hypertension points to restriction.

Where it is taught: Section C03, the Kussmaul sign paragraph, the restrictive cardiomyopathy paragraph, the comparison table, and the stop and predict box of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C03-RT01

Retrieval exercise: Rebuild the path from pericarditis to constriction

A competent response contains

· Step 1: a fibrinous, purulent, or caseous exudate, and purulent and caseous exudates are the most likely to take this path, as is radiation injury.

· Step 2: granulation tissue organizes the exudate.

· Step 3: the visceral and parietal layers fuse and obliterate the pericardial space, so the normal 15 to 50 mL of lubricating fluid is lost.

· Step 4: dense fibrosis forms, sometimes with calcification that can be seen on a plain film or CT.

· Step 5: a rigid shell fixes total cardiac volume, while the myocardium inside still relaxes normally.

· Organization of exudate replaces a compliant sac with a fixed shell, so the pressure-volume relationship loses its reserve volume and pressure rises steeply almost at once.

Distinctions that matter

· Granulation tissue organizing an exudate is ordinary repair by fibrosis; what makes it dangerous here is the space it fuses shut.

· The shell changes the sac, not the muscle, which is the whole basis for separating constriction from restriction.

Common misconceptions

· Expecting every pericarditis to constrict. Fibrinous and serofibrinous exudates from viral disease usually resolve without scarring.

· Thinking calcification is required for constriction. It is common and helpful on imaging but not necessary.

Where it is taught: Section C03, the flow from pericarditis to constriction, with the fluid volume from C01 of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C04-BD1

Brain dump 1 · foundational retrieval: Effusion: the curve, the causes, and the signs

A competent response contains

· A pericardial effusion is fluid in excess of the normal 15 to 50 mL.

· The fibrous pericardium has a J-shaped pressure-volume curve: over the small reserve volume added fluid barely raises pressure, and beyond it collagen fibers straighten and pressure climbs steeply.

· Fluid added over minutes reaches the steep limb at once, so as little as about 150 to 200 mL can compress the heart.

· Fluid collecting over weeks to months lets the sac stretch and remodel, shifting the curve to the right, so a liter or more can accumulate with little rise in pressure.

· Serous effusions are transudates from shifted Starling forces: raised venous pressure in heart failure, or low plasma oncotic pressure in nephrotic syndrome and cirrhosis.

· Hypothyroidism produces a large, slowly forming, protein-rich effusion through increased capillary leak and reduced lymphatic drainage.

· Exudates come from the pericarditis causes in C02; hemorrhagic fluid points to trauma, free wall rupture, aortic dissection, malignancy, tuberculosis, uremia, or a bleeding tendency.

· Findings and mechanisms: muffled distant heart sounds because fluid separates the heart from the chest wall and dampens sound; a rub that may fade because fluid separates the inflamed layers, though it can persist; low-voltage QRS, under 5 mm in all limb leads, because fluid attenuates the signal; electrical alternans because the heart swings within a large effusion and changes its axis on alternate beats; an

enlarged globular water bottle silhouette, usually needing more than about 250 mL to show; an echo-free space on echocardiography where fluid separates the layers.

· Low voltage is not specific: obesity, emphysema, hypothyroidism, and amyloid also cause it. Electrical alternans in this setting suggests a large effusion and possible tamponade.

Distinctions that matter

· Transudate against exudate is decided by mechanism: pressure or oncotic imbalance against increased permeability from inflammation.

· The signs of effusion report how much fluid is present; whether filling is impaired depends on pericardial pressure.

Common misconceptions

· Reading a large silhouette or low voltage as evidence of compression. Both describe fluid, not pressure.

· Expecting a small effusion to be harmless. Over minutes about 150 to 200 mL is enough to reach the steep limb.

Where it is taught: Section C04, first two paragraphs and the table of findings and their mechanism of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C04-BD2

Brain dump 2 · integration and application: Two patients, and why volume is the wrong number

A competent response contains

· Patient A sits far to the right on a curve that has shifted right: four months of slow stretch remodeled and enlarged the sac, so pericardial pressure stayed below right atrial pressure, about 2 to 6 mmHg, and filling continued despite 1,200 mL.

· Patient B sits on the steep limb of an unstretched curve: 200 mL in 20 minutes gave the sac no time to stretch, so pericardial pressure rose past right atrial and right ventricular diastolic pressure.

· The chamber with the lowest pressure stops filling first, which is the thin-walled low-pressure right heart, and output falls.

· Patient A is likely to show an enlarged globular water bottle silhouette on the chest film and low-voltage QRS complexes, possibly with electrical alternans, with distant heart sounds.

· Patient B has too little fluid to enlarge the silhouette, since more than about 250 mL is usually needed, and shows hypotension with distended neck veins.

· The lung fields are clear in patient A because left heart pressures are not raised, so a large chronic effusion can be well tolerated while still producing the effusion signs.

· The rule: the speed of accumulation predicts compression better than the volume, and judging by volume alone would have predicted the opposite outcome in this pair.

Distinctions that matter

· Position on the curve, not volume, decides whether pericardial pressure exceeds right atrial pressure.

· Signs of a large effusion and signs of compression are different sets: patient A has the first and patient B the second.

Common misconceptions

· Expecting the larger effusion to be the dangerous one.

· Expecting pulmonary edema with a large effusion. Left heart filling pressures are not raised, so the lung fields stay clear.

Where it is taught: Section C04, the example on why rate matters more than volume, with the findings table of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C04-PR01

Prediction exercise: Predict the effect of a slowly accumulating effusion

A competent response contains

· The curve shifts to the right, because slow stretch over months lets the sac remodel and enlarge its reserve volume.

· Pericardial pressure stays below right atrial pressure, about 2 to 6 mmHg, so the sac is still on the flat part of its curve.

· Diastolic filling continues and stroke volume is maintained, so blood pressure is normal.

· QRS amplitude falls, giving low voltage in the limb leads, because fluid between the heart and the electrodes attenuates the signal, and hypothyroidism itself also lowers voltage.

· The silhouette is enlarged and globular, the water bottle shape, since the volume is far above the roughly 250 mL needed to show.

· The lung fields are clear, because left heart filling pressures are not raised.

· The same volume in one hour would reach the steep limb of an unstretched curve, pericardial pressure would exceed right heart diastolic pressure, and tamponade with hypotension and distended neck veins would follow.

Distinctions that matter

· A large effusion and a high pericardial pressure are separate things, and only the second impairs filling.

· Low voltage here has two contributors, the fluid and the hypothyroidism, which is part of why the sign is not specific.

Common misconceptions

· Predicting tamponade from the volume alone.

· Expecting hypothyroid effusions to be thin transudates. They are protein-rich and form slowly through capillary leak and reduced lymphatic drainage.

Where it is taught: Section C04, the pressure-volume paragraph, the example, and the findings table of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C05-BD1

Brain dump 1 · foundational retrieval: Tamponade: definition, causes, and the fall in output

A competent response contains

· Tamponade is a hemodynamic state, not a fluid volume: it exists once pericardial pressure rises enough to impair filling.

· Any effusion can cause it; the classic acute cause is hemopericardium, and nontraumatic causes include malignancy, uremia, and pericarditis.

· Penetrating trauma often enters the right ventricle, the most anterior chamber.

· Free wall rupture after transmural infarction occurs most often about 3 to 7 days after the event, when macrophages have softened the necrotic wall, and causes sudden tamponade with pulseless electrical activity.

· Type A aortic dissection can rupture backward into the sac, because the proximal ascending aorta lies inside the pericardial reflection.

· Sequence: fluid or blood collects faster than the sac can stretch; pericardial pressure exceeds right heart diastolic pressure; the right atrium and ventricle collapse in diastole; diastolic pressures in all chambers equalize with pericardial pressure; preload and stroke volume fall with tachycardia compensating; hypotension, distended neck veins, and distant heart sounds follow.

· The thin-walled, low-pressure right heart is compressed first.

· Right atrial, right ventricular diastolic, pulmonary artery diastolic, and pulmonary capillary wedge pressures become nearly equal to pericardial pressure.

· Beck triad: hypotension from low stroke volume, distended neck veins from high venous pressure, and distant heart sounds from the fluid.

· Sinus tachycardia is the compensating response, and electrical alternans may appear.

Distinctions that matter

· Tamponade is defined by pressure, so a small rapid bleed can cause it and a large chronic effusion may not.

· Equalization of diastolic pressures is the hallmark of a shared tense space, and it happens in constriction as well.

Common misconceptions

· Naming a threshold volume for tamponade. There is none, because the rate of accumulation and the state of the sac decide it.

· Expecting the left ventricle to fail first. The low-pressure right heart is compressed first.

Where it is taught: Section C05, causes paragraph, the flow on rising pericardial pressure, and the equal pressures paragraph of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C05-BD2

Brain dump 2 · integration and application: Separate fluid under pressure from a fixed shell

A competent response contains

· In tamponade filling occurs mainly during ventricular systole, when ejection briefly lowers intrapericardial volume and pressure.

· The jugular x descent is preserved, because it coincides with that systolic fall in pericardial pressure; the y descent is blunted or absent, because pericardial pressure stays above right atrial pressure after the tricuspid valve opens, so early diastolic filling cannot occur.

· Comparison: onset acute or subacute against chronic; pulsus paradoxus usual against less common; Kussmaul sign usually absent against typical; y descent blunted or absent against steep; filling impaired throughout diastole against mid and late diastole only; diastolic pressures equalize in both; no extra heart sound against a pericardial knock.

· Fluid under pressure compresses the heart throughout diastole, while a shell allows rapid early filling and stops it only once the limit is reached.

· Kussmaul sign is usually absent in tamponade because the fluid still transmits the inspiratory fall in intrathoracic pressure to the right atrium.

· Pulsus paradoxus is measured by inflating the cuff above systolic pressure and deflating slowly during quiet breathing: the first pressure is where Korotkoff sounds appear in expiration only, the second is where they are heard with every beat in both phases, and the difference is the pulsus; more than 10 mmHg is abnormal.

· Mechanism: inspiration lowers intrathoracic pressure, venous return to the right ventricle increases, the right ventricle cannot expand outward against the tense sac, the septum bulges into the left ventricle, the pulmonary veins also hold more blood, left ventricular filling and stroke volume fall, and systolic pressure drops more than 10 mmHg.

· Other causes: severe asthma and COPD through large swings in intrathoracic pressure, and massive pulmonary embolism through right ventricular strain.

Distinctions that matter

· The y descent reports early diastolic filling, which is why it is lost with fluid and steep with a shell.

· Pulsus paradoxus and Kussmaul sign point in opposite directions between the two diseases, because fluid transmits intrathoracic pressure and a rigid shell does not.

Common misconceptions

· Treating pulsus paradoxus as specific for tamponade.

· Expecting Kussmaul sign in tamponade. The fluid still transmits the inspiratory pressure fall to the right atrium.

Where it is taught: Section C05, the paragraph on equal pressures and filling, the pulsus paradoxus flow and example, and the tamponade against constriction table of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C05-RT01

Retrieval exercise: Rebuild pulsus paradoxus

A competent response contains

· Step 1: inspiration lowers intrathoracic pressure.

· Step 2: venous return to the right ventricle increases.

· Step 3: the right ventricle cannot expand outward, because the tense sac fixes total cardiac volume; the principle linking the ventricles is ventricular interdependence.

· Step 4: the septum bulges into the left ventricle.

· Step 5: the pulmonary veins, which lie in the chest outside the sac, also hold more blood on inspiration.

· Step 6: left ventricular filling and stroke volume fall.

· Step 7: systolic pressure drops, and a fall of more than 10 mmHg is abnormal.

· Total cardiac volume is fixed through the respiratory cycle, so any gain in right heart filling becomes a loss in left heart filling.

Distinctions that matter

· Two things reduce left ventricular filling on inspiration: the septum shifting and blood held in the pulmonary veins.

· The normal small inspiratory fall in systolic pressure is present in everyone; interdependence inside a tense sac is what makes it large.

Common misconceptions

· Thinking the pulse disappears on expiration. The beats lost are the inspiratory ones, which carry the smallest left ventricular stroke volume.

· Expecting the right ventricle to expand outward into the sac. The sac is tense, so the only room available is across the septum.

Where it is taught: Section C05, the flow on pulsus paradoxus in tamponade and the example on measuring it of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C05-CA01

Clinical application: Sudden hypotension six days after an anterior infarction

A competent response contains

· Recognize: free wall rupture of the infarcted left ventricle has filled the sac with blood, producing acute cardiac tamponade with pulseless electrical activity; day 6 fits rupture, which occurs most often about 3 to 7 days after a transmural infarct, when macrophages have softened the necrotic wall.

· Explain: blood collects faster than the sac can stretch, pericardial pressure exceeds right heart diastolic pressure, the right atrium and right ventricle collapse in diastole, and right atrial, right ventricular diastolic, pulmonary artery diastolic, and pulmonary capillary wedge pressures equalize with pericardial pressure; the ventricles cannot fill, so stroke volume approaches zero while the conduction system keeps producing organized complexes, which is pulseless electrical activity.

· Differentiate: acute left ventricular failure would raise left-sided pressures and give pulmonary congestion with crackles rather than clear lung fields, and it would not show diastolic collapse of the right chambers; constrictive pericarditis is chronic, gives a pericardial knock and a steep y descent with Kussmaul sign, and the acute onset, pulsus paradoxus, and echo-free space here separate this patient from it.

· Investigate: on echocardiography look for an echo-free space around the heart and diastolic collapse of the right atrium and right ventricle; at the bedside look for Beck triad with hypotension, distended neck veins, and distant heart sounds, pulsus paradoxus above 10 mmHg, sinus tachycardia, and possible electrical alternans, with equal diastolic pressures across the chambers.

· Decide: the fibrous pericardium has a J-shaped pressure-volume curve with only a small reserve volume, and bleeding over minutes gave the sac no time to stretch, so the sac is on the steep limb of the curve where pressure climbs sharply, which is why as little as about 150 to 200 mL is enough.

· Justify: inspiration increases right ventricular filling, the right ventricle cannot expand against the tense sac, the septum bulges into the left ventricle, and the pulmonary veins hold more blood, so left ventricular filling and stroke volume fall and systolic pressure drops by more than 10 mmHg; the jugular venous pressure does not rise with inspiration here because fluid still transmits the inspiratory fall in intrathoracic pressure to the right atrium, while the rigid shell of constriction blocks that transmission and produces Kussmaul sign.

Distinctions that matter

· Clear lung fields with distended neck veins and hypotension separate compression of the whole heart from left ventricular pump failure.

· An organized rhythm with no pulse is a filling problem here, not a conduction problem.

Common misconceptions

· Assuming a large volume of blood is needed. Speed decides, and about 150 to 200 mL added acutely is enough.

· Expecting Kussmaul sign in any patient with raised venous pressure and impaired filling. It belongs to the rigid shell, not to fluid.

Where it is taught: Section C05, causes paragraph, both flows, the measurement example, the comparison table, and the stop and predict box of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C05-Q02

Board-style question: Blunted y descent in a tense sac

Correct answer: A. Pericardial pressure remains above right atrial pressure after the tricuspid valve opens, so early diastolic filling cannot occur

Why it is right: Pericardial pressure remains above right atrial pressure once the tricuspid valve opens, so there is no early diastolic emptying of the atrium and the y descent is lost, while the x descent persists because ejection briefly lowers intrapericardial volume and pressure during systole.

B is wrong: A rigid fibrocalcific pericardium permits very rapid early diastolic filling until the shell is reached, which produces a steep y descent and a pericardial knock, neither of which is present here.

C is wrong: Loss of atrial contraction would remove the a wave and the late diastolic contribution to filling rather than the y descent, and this patient is in sinus rhythm.

D is wrong: Impaired myocardial relaxation from amyloid infiltration is restrictive cardiomyopathy, which leaves the pericardium normal, raises left-sided pressure above right, and does not blunt the y descent.

A competent response contains

· Hypotension, distended neck veins, distant heart sounds, pulsus paradoxus above 10 mmHg, and equal diastolic pressures in a patient with malignancy indicate cardiac tamponade.

· Filling occurs mainly during ventricular systole, when ejection lowers intrapericardial volume and pressure, so the x descent survives and the y descent, which reports early diastolic filling, is lost.

Distinctions that matter

· A blunted or absent y descent belongs to tamponade; a steep y descent belongs to constriction.

Common misconceptions

· Reading the absent y descent as a myocardial relaxation problem. The ventricle would relax normally if pericardial pressure allowed it to fill.

Where it is taught: Section C05 of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C06-BD1

Brain dump 1 · foundational retrieval: How tumor reaches the heart and what it does there

A competent response contains

· Autopsy series find cardiac or pericardial metastases in roughly 5 to 10% of patients who die of cancer, and metastases are far more common than primary cardiac tumors such as myxoma.

· Sites in order of frequency: pericardium, then epicardium, then myocardium.

· Direct extension: lung carcinoma, breast carcinoma, esophageal carcinoma, mediastinal tumors.

· Retrograde lymphatic spread through mediastinal nodes: lung and breast carcinoma, lymphoma.

· Hematogenous seeding: melanoma, leukemia, lymphoma, sarcomas.

· Growth along the inferior vena cava into the right atrium: renal cell carcinoma, hepatocellular carcinoma.

· Proximity explains lung and breast, which lie next to the heart and are common, so they account for the largest numbers of cases; wide bloodstream spread explains melanoma and the hematologic cancers.

· Melanoma is less common but involves the heart in a high proportion of patients with widespread disease.

· Leukemia and lymphoma often infiltrate the heart.

Distinctions that matter

· Largest number of cases and highest proportion of patients involved are different measures: lung and breast lead the first, melanoma the second.

· Growth along the inferior vena cava is a route into a chamber rather than onto the serosa, which is why it causes obstruction rather than effusion.

Common misconceptions

· Thinking of myxoma first when a cardiac mass or pericardial effusion appears in a patient with cancer. Metastases greatly outnumber primary tumors.

· Assuming the myocardium is the usual site. The pericardium is the most frequent.

Where it is taught: Section C06, first paragraph and the table of routes of spread of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C06-BD2

Brain dump 2 · integration and application: The effusion that keeps coming back

A competent response contains

· Sequence: tumor reaches the pericardium by extension, lymphatics, or blood; serosal implants with fragile new vessels bleed; lymphatic obstruction blocks fluid drainage; a bloody exudate with malignant cells accumulates; pericardial pressure rises until filling fails.

· The two mechanisms are bleeding from fragile leaky tumor vessels on the serosa and blockage of the mediastinal lymphatics that normally drain pericardial fluid.

· The result is a hemorrhagic exudate, and finding malignant cells on cytology of the pericardial fluid confirms tumor as the cause.

· Effusion forms faster than it drains, so it tends to recur and progress, which is why a rapidly growing effusion or one that reaccumulates after drainage raises suspicion of tumor.

· Malignancy is one of the most common causes of nontraumatic tamponade.

· A pericardial problem in a patient with cancer is not always tumor: mediastinal radiation given for the same cancer can later cause radiation pericarditis and constriction, and tumor encasing the heart can behave like constrictive pericarditis.

· Patients usually present with progressive dyspnea, chest discomfort, and the effusion signs in C04.

· Myocardial deposits are less often symptomatic but can disturb conduction and cause arrhythmias, and intracavitary tumor can obstruct flow.

Distinctions that matter

· Tumor can limit filling two ways: fluid under pressure in the sac, or a tumor shell around the heart that imitates constriction.

· Reaccumulation after drainage is the feature that separates a malignant effusion from a self-limited inflammatory one.

Common misconceptions

· Assuming every effusion in a cancer patient is malignant. Radiation pericarditis from earlier treatment is also a cause.

· Expecting a transudate. Tumor vessels bleed and lymphatics are blocked, so the fluid is a hemorrhagic exudate.

Where it is taught: Section C06, the paragraphs on pericardial deposits and clinical picture, the flow to tamponade, and the final paragraph of the CV-18-F Pericardial Disease notes.